

Test Scenarios

1.6.1 Positive Scenarios

- **TS_01_A - E2E Flow: Bürgeramt**
 - **Description:** Execution of a positive end-to-end scenario from Step 1 through Step 5. Selecting the "Bürgeramt" department, exactly 1 service, a location in Homburg, and the first available time slot. Step 5 is completed with valid personal data.
 - **Expected Result:** All steps up to the 5th are completed successfully; the "Reservieren" button transitions to the "enabled" state to proceed to Step 6.
- **TS_01_B - E2E Flow: Fahrerlaubnis**
 - **Description:** Execution of a positive end-to-end scenario from Step 1 through Step 5. Selecting the "Fahrerlaubnis" department, exactly 1 service, a location in Homburg, and the first available time slot. Step 5 is completed with valid personal data.
 - **Expected Result:** All steps up to the 5th are completed successfully; the "Reservieren" button transitions to the "enabled" state to proceed to Step 6.
- **TS_02 - UI: Service Counter Limits**
 - **Description:** Testing the business logic for service selection at Step 2. Verifying counter accuracy, "Weiter" button activation, and compliance with the system-wide limit (max 4 services). The test concludes at Step 2.
 - **Test Scenarios:**
 - **Initial State:** Verify that the "Weiter" button is in the "disabled" state when all counters are 0.
 - **Activation (0 → 1):** Increase the counter for any service to 1. Verification: "Weiter" button becomes "enabled", and the (-) button becomes active.
 - **Reset (1 → 0):** Decrease the service counter to 0. Verification: (-) button becomes "disabled", and the "Weiter" button reverts to "disabled".
 - **Reaching the Limit (3 → 4):** Increase the total count of services to 4. Verification: All (+) buttons on the page are automatically locked ("disabled"), while (-) buttons and "Weiter" remain active.
 - **Recovery from Limit (4 → 3):** Click the (-) button when 4 services are selected. Verification: All (+) buttons on the page are unlocked ("enabled"), and the "Weiter" button remains active.
 - **Expected Result:** All interface elements (counters, +/- buttons, "Weiter" button) synchronously toggle their states ("enabled"/"disabled") in strict accordance with the number of selected services, preventing exceeding limits or submitting an empty booking.
- **TS_03 - UI: Filter Interactivity**
 - **Description:** Proceeding to Step 4. Verifying accordion expansion, default date values, and the activation of time/day-of-the-week checkboxes. Testing filter application with dynamic availability processing. The test concludes at Step 4 or proceeds to Step 5 depending on appointment availability.

- **Expected Result:**
 - The "Vorschläge filtern" accordion expands successfully; date range fields are pre-populated with the current date (From) and the date 21 days later (To), adjusted to the department's working days.
 - Time interval and day-of-the-week checkboxes are interactive and toggle to the "checked" state upon clicking.
 - **Outcome A (Slots Found):** The time grid is displayed; clicking the first available slot triggers the "Hinweis" modal window, and successful confirmation redirects the user to Step 5.
 - **Outcome B (No Slots Found):** A valid system message "Kein freier Termin verfügbar" (No free appointments available) is displayed; the test concludes at Step 4.
- **TS_04 - Navigation: Backward & Cache Stability**
 - **Description:** Verifying cache stability, preservation of the selected time slot, and dynamic data updates when navigating backward to Step 1.
 - **Expected Result:**
 - **Return to Step 4:** Navigating back from Step 5 to Step 4 retains: Funktionseinheit, Anliegen, Standort, and Termin. The Übersicht panel displays all four fields correctly.
 - **Return to Step 3:** Navigating back from Step 4 to Step 3 retains: Funktionseinheit, Standort, and Termin. The "Anliegen" data is cleared from the Übersicht panel.
 - **Return to Step 2:** Navigating back to Step 2 retains only: Funktionseinheit. All subsequent selections (Anliegen, Standort, Termin) are cleared from the Übersicht panel.
- **TS_14 - UI: Global Header & Overview Consistency**
 - **Description:** Verifying the stability and correct rendering of global interface elements (Header, Footer).
 - **Expected Result:**
 - **Header:** The logo and two icons are displayed on all steps (1–6); elements remain present and accessible throughout every step of the process.
 - **Footer:** All links (Hilfe, Impressum, Datenschutz, etc.) trigger modal windows upon clicking. The "Schließen" (Close) button within each window correctly closes the popup.
- **TS_15 - UI: Dynamic Übersicht & Step Indicator**
 - **Description:** Verifying the correctness of the dynamic "Übersicht" (Summary) block, data accuracy within it, and the synchronization of the Step Indicator with the current booking stage.
 - **Expected Result:**
 - **Step Indicator:** The progress indicator (1–5) visually highlights the currently active step corresponding to the page content.
 - **Übersicht (Dynamics):**
 - **Step 1:** The "Übersicht" block is hidden.
 - **Steps 2–5:** The block is present with the accordion expanded, displaying accurate information regarding selected services, location, and time.

1.6.2 Negative & Edge Case Scenarios

- **TS_06 - Filter: No Available Slots Handling**
 - **Status:** Blocked by Test Data. The test scenario is 80% automated (validation of the "Kein freier Termin" message).
 - **Reason:** Full implementation (filter reset + continuation) is blocked due to the inability to predict slot availability after filtering in the current test environment.
 - **Description:** A scenario where filter parameters at Step 4 return an empty result (no slots available). Verification of message display and filter reset via automated testing to continue the scenario.
 - **Expected Result:**
 - The calendar grid is hidden, and the message "Kein freier Termin verfügbar" (No free appointments available) is displayed. The filter is collapsed, and the "Zurücksetzen" (Reset) button is clickable.
 - The automated test clears the selected filter checkboxes, re-clicks "Filtern" (Filter), after which available slots appear.
 - The test clicks the first available slot, selects "Ja" (Yes) in the confirmation window, and proceeds to Step 5.
- **TS_07 - Security: Deep Linking Protection**
 - **Description:** Attempting a direct URL navigation to an internal booking step (Step 5) without an active session.
 - **Expected Result:** The system blocks access to the form and displays an error message.
- **TS_08 - Security: Form Validation**
 - **Description:** Verification of negative validation, input field security, and the state of the "Reservieren" (Reserve) button at Step 5 for both "Bürgeramt" and "Fahrerlaubnis" processes.
 - **Expected Result:**
 - **Real-time Feedback:** Upon entering invalid data (e.g., incorrect email format, invalid year, or mismatching emails), the system must immediately trigger a visual error response:
 - **Error Message:** Display of a specific red error text (e.g., "Die Mailadressen stimmen nicht überein").
 - **Visual Indicator:** The field/icon status changes from "valid" (green) to "invalid" (red).
 - **Button State Logic:** * The "Reservieren" button remains strictly disabled (`disabled` attribute / `aria-disabled="true"`) while mandatory fields are empty or contain invalid data.
 - The button becomes enabled **only after** all validation checks for mandatory fields are passed and the consent checkbox is checked.
- **TS_16 - System: Paste Restriction on Email Confirmation**
 - **Description:** Verifying that the email confirmation field ("Email wiederholen") strictly requires manual input by blocking the paste (Ctrl+V/Right-click) functionality.
 - **Expected Result:** The application must prevent any text from being pasted into the email confirmation field. The field must remain empty or unchanged

upon a paste attempt, ensuring the user is forced to type the input manually. This restriction acts as a primary preventative measure before secondary validation (email mismatch) is triggered.

- **TS_09 - Session: Warning & Extension**

- **Description:** Simulating browser clock progression via Playwright's `page.clock` to reach the 01:00 session threshold at Step 5, verifying client-side warning modal interaction and state persistence.
- **Expected Result:**
 - When the client-side session timer approaches the critical threshold (1 minute remaining), an "Ihre Sitzung läuft aus in 1 Minute" warning triggers on Step 5 and the warning modal opens.
 - Clicking the "Schliessen" button closes the modal window without resetting form inputs.
 - *Note:* This scenario validates client-side UI timer logic, warning overlay behavior, and form data retention under faked browser time. Server-side session renewal is outside the scope of virtual clock testing.

- **TS_10 - Session: Timeout Expiration**

- **Description / Precondition:** The user is on Step 5 with a partially filled form. No interface interactions occur until the session timer (25 minutes) fully expires.
- **Expected Result:** An automatic redirect to Step 1 occurs with a session reset parameter (URL: `.../homburg/?rs`). The Step 5 interface is completely locked, entered data is erased, and the footer timer restarts from the initial 25 minutes (displaying 24 on-screen).

- **TS_11 - System: Network Drop Simulation**

- **Status:** Deferred.
- **Reason:** Risk of data integrity violations in the live system. Forcing a connection drop during the booking submission process may result in "phantom" records or orphaned sessions in the database.
- **Description:** The user is on Step 5 and has completed the form. At this moment, the browser context is switched to "Offline" mode. An attempt is made to submit the form or interact with elements requiring server communication.
- **Expected Result:** The interface remains stable, and data in the DOM is preserved. Upon attempting to submit the form, the system displays a valid notification regarding the lost network connection (or the browser generates a standard network availability error, blocking the submission of "stale" data).

- **TS_12 - System: Parallel Booking Conflict**

- **Status:** Deferred.
- **Reason:** Test environment limitations (dependency on a shared backend). Executing parallel booking requests requires an isolated database with rollback capabilities after each run, which is not feasible without backend code intervention.
- **Description:** Testing concurrent access to a single time slot by simultaneously sending requests from Step 5 to Step 6 using two independent Playwright instances.

- **Expected Result:** The first request completes successfully (booking confirmed), while the second thread intercepts an API error (Conflict 409), and the interface displays a validation message to the user indicating that the selected slot is already occupied.
- **TS_13 - System: API 500 Error Handling**
 - **Description:** Verifying interface behavior when critical calendar data is unavailable (simulating an HTTP 500 error for `calendar.js` during transition to Step 4).
 - **Expected Result:** The system must handle the API 500 error gracefully. When `calendar.js` fails with HTTP 500 during transition from Step 3 (Location) to Step 4 (Date selection), rendering of Step 4 must be blocked, and the system should display a clear error notification ("Fehler bei der Terminauswahl") instead of rendering an unhandled blank screen.
- **TS_17 - Error Message on Email Mismatch**
 - **Description:** Validation of the email confirmation logic by entering non-matching email addresses in the "E-Mail" and "E-Mail wiederholen" fields. The scenario proceeds through steps 1–4, then attempts to trigger validation by entering distinct email strings and shifting focus.
 - **Expected Result:** The system identifies the inconsistency and displays the specific error message: "Die Mailadressen stimmen nicht überein." in the designated error container.
- **TS_18 - System: Field Validation on Incomplete Submission**
 - **Description:** Verification of the form submission state when required fields in Step 5 are left incomplete. The scenario completes steps 1–4, navigates to step 5, and populates all mandatory personal data fields except for one (e.g., leaving the "Nachname" field blank).
 - **Expected Result:** The "Reservieren" button remains in a "disabled" state, preventing the user from proceeding to Step 6, as the validation requirements for mandatory fields are not met.